



UNIVERSITY OF RUHUNA

Faculty of Engineering

End-Semester 8 Examination in Engineering: October 2024

Module Number: EE8209

Module Name: Information Security

[Three Hours]

[Answer all questions, each question carries 10 marks]

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- Q1 a) Why is it crucial that cryptographic algorithms be created with the expectation that the public would be aware of them?
[1.0 Mark]
- b) How can a man-in-the-middle attack compromise the integrity of data during transmission?
[1.0 Mark]
- c) Use the following decryption algorithm to decrypt the given cipher text.
$$\text{Plaintext letter} = (\text{Ciphertext letter} - \text{Key letter}) \bmod 26$$

Note that modulus operation applies to each letter. Use the following mapping.
$$A = 0, B = 1, C = 2, \dots Z = 25$$
- i) Find the plaintext, if the Cipher text = MIOGV and Key = LIVEO.
[2.0 Marks]
- ii) Briefly explain two (2) security issues associated with the above cryptographic method.
[1.0 Mark]
- iii) Discuss the concepts of confusion and diffusion in the context of the above cryptographic system.
[1.0 Mark]
- d) Use Double Transposition Cipher to encrypt the following plain text. The key is [2413], [3142]. Hint: The order of the key is [ROW], [COLUMN].
- i) Derive the cipher text, if the plain text = "MEETMEATTHESHOP".
[2.0 Marks]
- ii) Indicate whether confusion and/or diffusion are achieved by the cryptosystem mentioned above.
[1.0 Mark]
- e) Discuss the security features and vulnerabilities of the Enigma machine used by the German military during World War II.
[1.0 Mark]
- Q2 a) List two (2) advantages of block ciphers over stream ciphers.
[1.0 Mark]

- b) Consider the following Feistel cipher design with *three* rounds and the plain text, $P = (L_0, R_0)$.

Encryption:

$$\begin{aligned} L_i &= R_{i-1} \\ R_i &= L_{i-1} \oplus F(R_{i-1}, K_i) \end{aligned}$$

The cipher text is the resulting text at the end of three rounds, i.e., $C = (L_3, R_3)$. Here, F is the round function and K_i is the subkey for round i .

- i) Given, $P = 1101\ 0110$, $K_1 = 1010$, $K_2 = 1010$, and $K_3 = 1010$, compute the final cipher text if $F(R_{i-1}, K_i) = R_{i-1} \oplus K_i$.
[2.5 Marks]
- ii) Comment on the security aspects of using the same sub-key in each round.
[1.0 Mark]
- c) In cryptography, encryption is modelled as $C = E(P, K)$ and decryption is modelled as $P = D(C, K)$, where C is the cipher text, P is the plaintext, K is the key, E denotes encryption, and D denotes decryption.
 - i) Write the corresponding model equations for encryption and decryption in ECB mode?
[1.0 Mark]
 - ii) What is the main issue associated with the ECB mode?
[0.5 Marks]
- d) Message Authentication Code (MAC) is used to check integrity of messages in symmetric key cryptography.
 - i) Explain using notations how the MAC can be used ensure integrity, if a plaintext block (P_i) is changed as X by Trudy.
[2.0 Marks]
 - ii) Why is it recommended to use a different key to generate the MAC?
[1.0 Mark]
- e) List one (1) advantage and one (1) disadvantage symmetric key encryption has over public key encryption.
[1.0 Mark]

- Q3 a) Following are the steps executed in the RSA encryption algorithm.

- Two prime numbers are selected as p and q
- $N = pq$
- Compute the totient function, $\phi(N) = (p - 1)(q - 1)$
- Select e such that, e is relatively prime with $\phi(N)$. i.e., $\gcd[e, \phi(N)] = 1$
- $d = e^{-1} \bmod \phi(N)$
- Encryption: $C = M^e \bmod N$
- Decryption: $M = C^d \bmod N$

- i) What actions are necessary to crack the RSA cryptosystem?
[1.0 Mark]
- ii) For $p = 5$ and $q = 11$, calculate the private and public key values, and derive the cipher text for plaintext, $M = 10$. Show your workings. Select e as 3.
[2.0 Marks]

iii) Decipher the cipher text, $C = 10$, and find out the corresponding plaintext. [1.5 Marks]

b) Alice and Bob want to securely exchange a shared secret key using the Diffie-Hellman key exchange protocol. They agree on the following public parameters:

Prime number $(p) = 23$

Generator $(g) = 5$

i) Alice selects her private key, $a = 4$ and Bob selects his private key, $b = 3$. Show with working, how the shared secret is computed. [2.0 Marks]

ii) Briefly explain why someone listening to the channel cannot generate the shared secret key without having access to a or b . [1.0 Mark]

c) Given the public key K_{pub} and the private key K_{pvt} of a user, describe the meaning of the following operations.

i) $C = \{M\}_{K_{pub}}$

ii) $S = [M]_{K_{pvt}}$

iii) $\{H(M)\}_{K_{pub}}$ [3x0.5 Marks]

d) State one (1) advantage and one (1) disadvantage of Elliptic Curve Cryptography (ECC). [1.0 Mark]

Q4 a) Briefly explain why collision resistance is a crucial property for a hash function in cryptographic applications. [1.0 Mark]

b) Alice's computer needs to have access to a symmetric key K_A . Consider the following method and answer the question.

The key is $K_A = h(\text{Alice's password})$, and K_A is not stored on Alice's computer. Whenever K_A is required, Alice enters her password, and the key is generated. Discuss one (1) advantage and disadvantage of this method. [2.0 Marks]

c) Hash based Message Authentication Code (HMAC) is important in preserving integrity of data.

i) Briefly explain why it is important to embed the key in HMAC. [1.0 Mark]

ii) Describe two (2) approaches where key can be embedded with the message in HMAC using notations. [1.0 Mark]

d) Consider the SSL protocol in Figure Q4. This protocol does not allow Bob to remain anonymous, since his certificate identifies him. Modify the protocol so that Bob can remain anonymous.

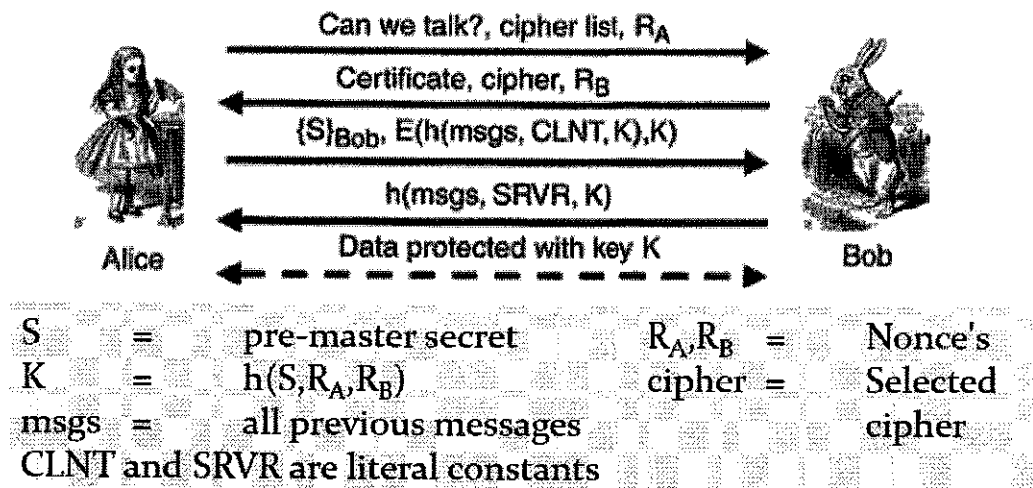


Figure Q4: Simplified SSL protocol

- [2.0 Marks]
- e) What is the main difference between the Main mode and Aggressive mode in IPSec protocol?
- [1.0 Mark]
- f) Explain how you would use text steganography to send a secret message to your boyfriend or girlfriend. Describe how you embed the message within a normal-looking text and outline how the recipient would decode it.
- [1.0 Mark]
- g) List two (2) watermarks you have seen in the real-world and mention their watermark type(s).

[1.0 Mark]

- Q5 a) Consider the mutual authentication protocol shown in Figure Q5, where K_{AB} is the shared symmetric key. Is there a way for Trudy to break this authentication mechanism and impose as Alice? Discuss your answer with reasoning.

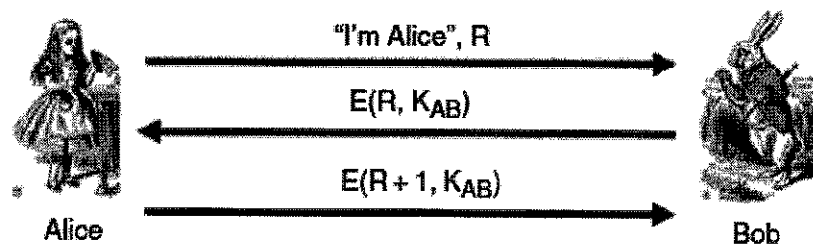


Figure Q5: A mutual authentication protocol

- [2.0 Marks]
- b) State three (3) best practices in creating and managing secure passwords.
- [1.5 Marks]
- c) Briefly explain why passwords or one-time passwords (OTPs) are generally unsuitable for authentication between two remote parties.

[1.0 Mark]

d) What is Equal Error Rate (EER) in biometrics?

[0.5 Marks]

e) Consider the following use case.

- Manager can read and write both Document1 and Document2, and has print permission on the Printer. However, the Manager does not have any access to the Database.
- Employee1 has read permission on Document1 and execute permission on the Database. Employee1 cannot print to the Printer.
- Employee2 has read and write permissions on Document2 and can execute operations on the Database but has no access to the Printer.
- Admin has full permissions on all objects.

Draw the Lampson's Access Control matrix for the above use case. Consider the necessary permissions as: Read (R), Write (W), Execute (E), Delete (D), Print (P).

[2.0 Marks]

f) Explain the difference between an Encrypted virus, Polymorphic virus and a Metamorphic virus.

[1.5 Marks]

g) Discuss the challenges of false positives and false negatives in Intrusion Detection Systems (IDSs). How do they affect the performance and reliability of an IDS.

[1.5 Marks]